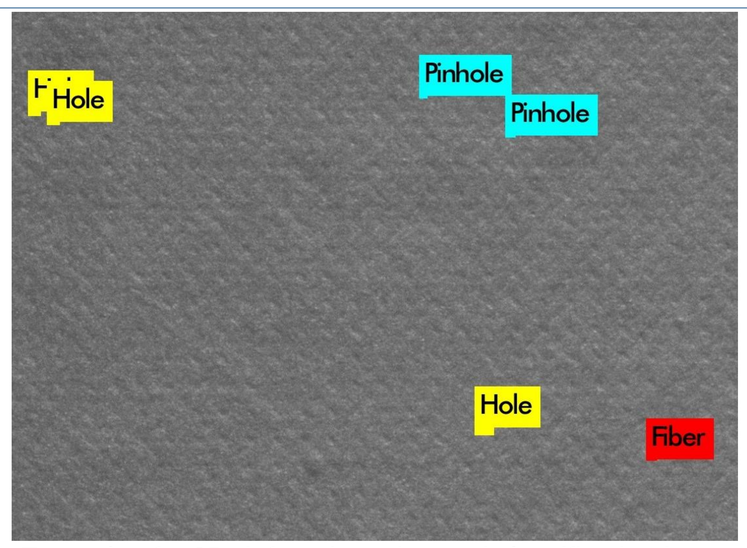
Mass-manufacturing and quality assurance solutions in SOFC value chain in FCH JU projects qSOFC and INNO-SOFC

Cells

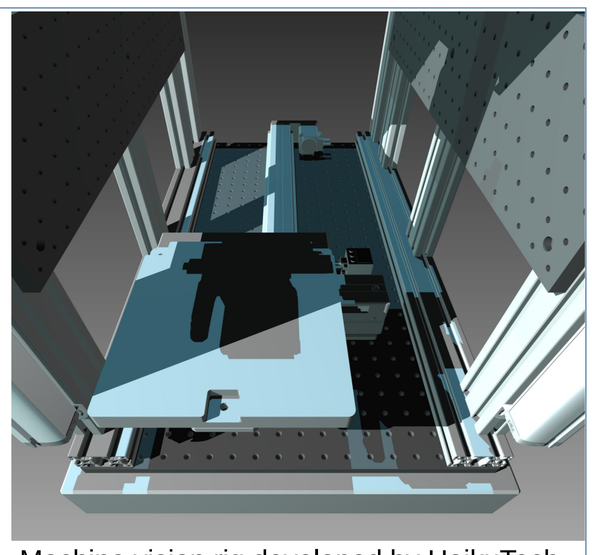
Targets
- Reduce cell manufacturing cost to 400 €/kW at production volume of 10 MW/year
- Increase production yield in all parts of stack manufacturing value chain to above 95% by automation and quality assurance

Methods
- Automated machine vision inspection system to detect faults during cell manufacturing
- Optimization of the cell manufacturing for high-speed manufacturing

Results
- Higher and more homogeneous cell quality
- Higher yield and lower scrap rate throughout production



Example of a 3D defect detection and classification by the automated machine vision system in qSOFC

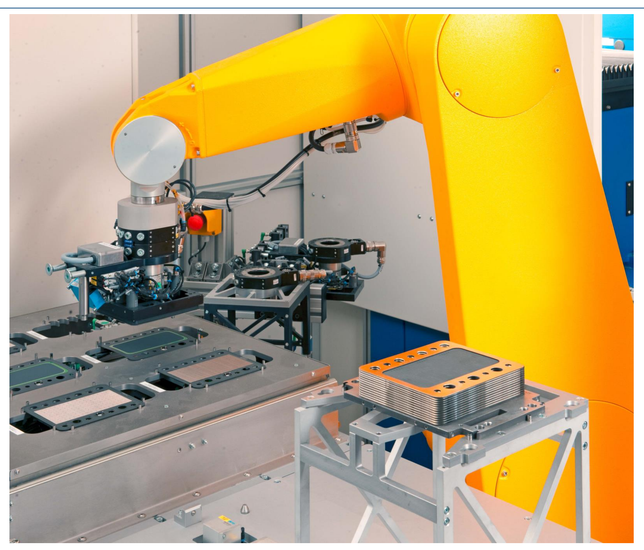

Machine vision rig developed by HaikuTech

Interconnects

Targets
- Interconnect manufacturing yield of 95%
- Development and validation of relevant quality assurance procedures in the interconnect manufacturing process

Methods
- Failure mode and effect analysis (FMEA) of interconnect manufacturing process
- Automated laser-scanning of interconnect geometries
- Integrated leak tightness measurement system to check interconnect welds/brazes

Results
- Improved quality of interconnects
- Reduced scrap rate


Example of stack assembly robot

Example of fully automated assembly and testing line

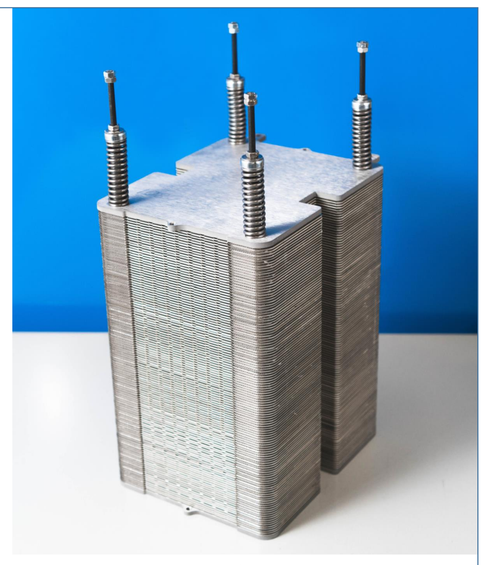
Stacks

Targets
- Streamline stack manufacturing process and cut cost by increasing yield and lowering OPEX and CAPEX related to manufacturing infrastructure
- Manufacture stacks for 60 kW INNO-SOFC system

Methods
- Testing and analysis of three different manufacturing and conditioning procedures using 30 stacks
- Statistical analysis of data

Results
- Optimized stack assembly and conditioning processes identified
- Potential cost-reductions corresponding to each process identified

Manufacturing process at Elcogen


INNO-SOFC stack

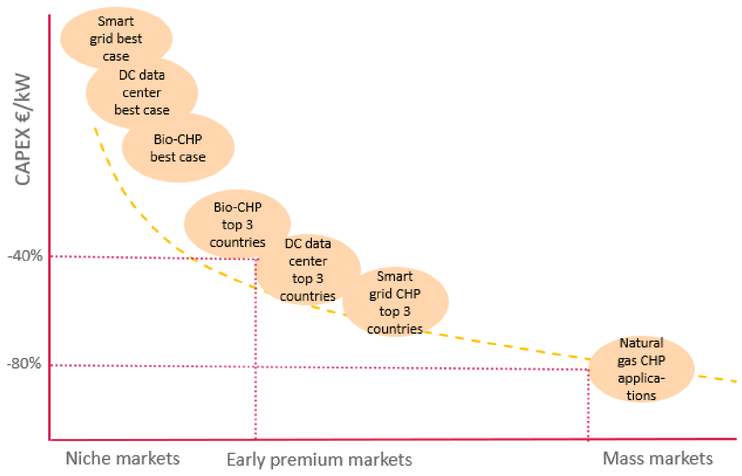
Systems

Targets
- System electrical efficiency 60%
- System total efficiency 85%
- High emphasis on cost reduction, 4000 € / kW system cost

Methods
- Effective collaboration throughout the value chain enables optimization of the whole system and its components. Key components are specifically designed and optimized for this system.

Results
- System conceptual design ready, exceeds project targets
- System design enables significant reduction in parts count and assembly time

Convion system



Use cases of Convion C50 SOFC

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